Brutal Series B09 - Heat | Acids, Bases & Salts | Integers | Rational Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A steel rod is 150 cm long at 25 deg C. It gets 3 mm longer for every 15 deg C rise in temperature. What is its length when it is heated to 100 deg C? [Heat]

- **(A)** 165 cm
- **(B)** 150.15 cm
- **(C)** 150.3 cm
- **(D)** 151.5 cm
- **(E)** 150.45 cm
- **(F)** 151.2 cm

2. Find the value of ((-15) + 24) - (-9) + (-30). [Integers]

- **(A)** -48
- **(B)** 12
- **(C)** -6
- **(D)** -12
- **(E)** 0
- **(F)** -30
- **(G)** 48

3. A scientist finds that 4 drops of acid exactly neutralise 5 mL of a base. How many drops of the same acid are needed to neutralise 20 mL of that base? [Acids, Bases & Salts]

- **(A)** 20 drops
- **(B)** 80 drops
- **(C)** 5 drops
- **(D)** 25 drops
- **(E)** 16 drops
- **(F)** 4 drops
- **(G)** 100 drops

4. Add the two rational numbers 3/4 and (-5/6), then simplify the result. [Rational Numbers]

- **(A)** -1/12
- **(B)** -8/12
- **(C)** 2/24
- **(D)** -1/2
- **(E)** 1/12
- **(F)** -2/10

5. A black water tank and a white water tank both start at 24 deg C in the sun. By noon the black tank has risen by 15 deg C and the white tank by 6 deg C. How much hotter is the black tank than the white tank at noon? [Heat]

- **(A)** 6 deg C
- **(B)** 69 deg C
- **(C)** 21 deg C
- **(D)** 30 deg C
- **(E)** 15 deg C
- **(F)** 9 deg C
- **(G)** 39 deg C

6. Find the value of (-8) x (-7) x (-2). [Integers]

- **(A)** 112
- **(B)** -128
- **(C)** -112
- **(D)** 56
- **(E)** 17
- **(F)** -17

7. A liquid turns blue litmus paper red but has no effect on red litmus paper. What kind of substance is it? [Acids, Bases & Salts]

- **(A)** Soap solution
- **(B)** A salt only
- **(C)** A base
- **(D)** A neutral substance
- **(E)** Distilled water
- **(F)** An acid

8. Find the value of (-2/3) - (-5/9). [Rational Numbers]

- **(A)** 3/12
- **(B)** -7/9
- **(C)** -3/9
- **(D)** 7/9
- **(E)** 1/9
- **(F)** -11/9
- **(G)** -1/9

9. A clinical thermometer is marked from 35 deg C to 42 deg C, with a small line every 0.2 deg C. How many small divisions are there in total over the whole scale? [Heat]

- **(A)** 14
- **(B)** 30
- **(C)** 35
- **(D)** 70
- **(E)** 36
- **(F)** 42

10. In a quiz, +5 marks are given for each correct answer and -3 marks for each wrong answer. A student attempts all 20 questions and gets 14 correct. What is the total score? [Integers]

(A) 88
(B) 58
(C) 70
(D) 34
(E) 52
(F) 46
(G) 40

11. A turmeric stain on a white cloth turns red when soap is rubbed on it. A few drops of lemon juice are then added and the red colour fades back to yellow. What does this show? [Acids, Bases & Salts]

(A) Both soap and lemon juice are acids
(B) Lemon juice is a base that turns turmeric red
(C) Soap is basic and lemon juice (acid) neutralises it
(D) Both soap and lemon juice are bases
(E) Soap is acidic and lemon juice is basic
(F) Turmeric is an acid

12. Find the value of (-4/7) x (14 / -8). [Rational Numbers]

(A) -56/56
(B) -7/4
(C) -1
(D) 7/4
(E) 2
(F) 1

13. On a sunny day, a black metal sheet warms up by 4 deg C every minute, while a white sheet warms up by 1.5 deg C every minute. Both start at 30 deg C. After 5 minutes, how much hotter is the black sheet than the white sheet? [Heat]

(A) 27.5 deg C
(B) 57.5 deg C
(C) 2.5 deg C
(D) 20 deg C
(E) 25 deg C
(F) 12.5 deg C
(G) 7.5 deg C

14. A diver starts at the water surface (height 0). She goes down 18 m, then rises 7 m, then goes down another 12 m. Her final position relative to the surface is: [Integers]

(A) -29 m
(B) -11 m
(C) 37 m
(D) -37 m
(E) -13 m
(F) -23 m

15. A patient has 21 mL of extra acid in the stomach. One antacid tablet can neutralise 3 mL of this acid. How many tablets are needed to neutralise all the extra acid? [Acids, Bases & Salts]

(A) 6 tablets
(B) 21 tablets
(C) 18 tablets
(D) 9 tablets
(E) 3 tablets
(F) 63 tablets
(G) 7 tablets

16. Find the value of (-3/5) divided by (9 / -10). [Rational Numbers]

(A) -2/3
(B) -27/50
(C) 6/5
(D) 27/50
(E) 2/3
(F) -3/2
(G) 3/2

17. At a beach on a hot afternoon, the land is at 41 deg C and the sea is at 29 deg C. By how many degrees is the land hotter, and which way does the breeze blow? [Heat]

(A) Sea is 12 deg C hotter; breeze blows from land to sea
(B) Land is 12 deg C cooler; breeze blows from land to sea
(C) Land is 29 deg C hotter; breeze blows from sea to land
(D) Land is 70 deg C hotter; breeze blows from sea to land
(E) Land is 12 deg C hotter; breeze blows from land to sea
(F) Both are equal; no breeze blows
(G) Land is 12 deg C hotter; breeze blows from sea to land

18. At dawn the temperature was -7 deg C. By noon it had risen by 9 deg C, and by night it had fallen by 5 deg C from the noon value. What was the temperature at night? [Integers]

(A) -3 deg C
(B) 7 deg C
(C) 3 deg C
(D) -21 deg C
(E) 2 deg C
(F) 11 deg C

19. An ant sting injects formic acid into the skin and causes burning. Which household substance should be rubbed on the sting to get relief, and why? [Acids, Bases & Salts]

(A) Vinegar, because it is a base that neutralises the acid
(B) Lemon juice, because it is a base
(C) Common salt, because it is neutral
(D) Lemon juice, because it is an acid that neutralises the sting
(E) Sugar, because it is a base
(F) Baking soda, because it is a mild base that neutralises the acid

20. Which of the following two rational numbers is greater: -3/4 or -5/7? [Rational Numbers]

(A) 5/7
(B) 3/4
(C) -3/4
(D) -3/7
(E) They are equal
(F) -5/7
(G) -5/4

21. Heat travels along a metal rod by conduction, moving 2.5 cm every 5 seconds. A drop of wax is stuck 40 cm from the heated end. After how many seconds does the wax there start to melt? [Heat]

(A) 200 seconds
(B) 160 seconds
(C) 80 seconds
(D) 32 seconds
(E) 16 seconds
(F) 100 seconds
(G) 50 seconds

22. Find the value of (-144) divided by 12, then that result divided by (-3). [Integers]

(A) 12
(B) 36
(C) -4
(D) -12
(E) 4
(F) -36
(G) 4032

23. Three liquids are each tested with BLUE litmus paper: vinegar turns it red, baking-soda solution causes no change, and lemon juice turns it red. How many of the three liquids are acids? [Acids, Bases & Salts]

(A) 3
(B) 1
(C) 2.5
(D) 4
(E) Cannot be decided
(F) 2
(G) 0

24. Find a rational number that lies exactly halfway between 1/3 and 1/2. [Rational Numbers]

(A) 3/5
(B) 2/5
(C) 5/6
(D) 5/12
(E) 1/6
(F) 7/12
(G) 2/12

25. A lab thermometer is first dipped in an ice-and-salt mixture and reads -8 deg C. It is then dipped in warm water and reads 47 deg C. By how many degrees did the reading rise? [Heat]

- **(A)** 8 deg C
- **(B)** 55 deg C
- **(C)** 39 deg C
- **(D)** 47 deg C
- **(E)** 54 deg C
- **(F)** 5.5 deg C

26. Find the value of ((-3) x 8) + (45 divided by (-9)). [Integers]

- **(A)** -24
- **(B)** -5
- **(C)** -29
- **(D)** -21
- **(E)** 19
- **(F)** 29
- **(G)** -19

27. When hydrochloric acid reacts with sodium hydroxide (a base), the reaction is a neutralisation. What are the products formed? [Acids, Bases & Salts]

- **(A)** Only water and no salt
- **(B)** A new acid and a new base
- **(C)** Hydrogen gas and water
- **(D)** Only a salt and no water
- **(E)** A salt (sodium chloride) and water
- **(F)** Two different salts
- **(G)** A salt and oxygen gas

28. Find the value of 5/8 + (-7/12). [Rational Numbers]

- **(A)** -12/20
- **(B)** 2/24
- **(C)** 1/12
- **(D)** -2/4
- **(E)** 12/20
- **(F)** 1/24

29. A metal ball just fits through a metal ring at 20 deg C, when the ball diameter is 50.0 mm. The ball expands by 0.01 mm for each 1 deg C rise. The ring hole stays 50.3 mm. At what temperature will the ball no longer pass through the ring? [Heat]

- **(A)** 20 deg C
- **(B)** 53 deg C
- **(C)** 23 deg C
- **(D)** 50 deg C
- **(E)** 300 deg C
- **(F)** 70 deg C
- **(G)** 30 deg C

30. Find the value of x if (-6) x x = 54. [Integers]

- **(A)** 9
- **(B)** -324
- **(C)** 324
- **(D)** 60
- **(E)** -9
- **(F)** -48
- **(G)** 48

31. A farmer field is too acidic and crops grow poorly. Which substance should be added to the soil to correct it, and what does it do? [Acids, Bases & Salts]

- **(A)** More acidic fertiliser, to balance the soil
- **(B)** Sugar, because it neutralises the acid
- **(C)** Lime (an acid), because it adds more acid
- **(D)** Common salt, because it is neutral
- **(E)** Lemon juice, because it makes the soil more basic
- **(F)** Lime (a base), because it neutralises the excess acid in the soil

32. Find the value of (-2/9) x (-3/4) x (6 / -5). [Rational Numbers]

- **(A)** -1/5
- **(B)** -1/6
- **(C)** -5/1
- **(D)** -6/30
- **(E)** 1/6
- **(F)** 1/5

33. At a hill station the temperature at 6 a.m. was 4 deg C. It then rose by 2 deg C every hour until noon. What was the temperature at 12 noon? [Heat]

- **(A)** 14 deg C
- **(B)** 18 deg C
- **(C)** 16 deg C
- **(D)** 28 deg C
- **(E)** 12 deg C
- **(F)** 10 deg C

34. Find the value of 25 + (-13) - 8 - (-20). [Integers]

- **(A)** 24
- **(B)** 50
- **(C)** 26
- **(D)** 0
- **(E)** -16
- **(F)** 4
- **(G)** -24

35. A wasp sting is alkaline (basic) and stings painfully. Which household item gives relief, and why? [Acids, Bases & Salts]

- **(A)** Vinegar, because it is a base
- **(B)** Soap solution, because it is a strong acid
- **(C)** Vinegar, because it is a mild acid that neutralises the base
- **(D)** Lime water, because it is an acid
- **(E)** Common salt, because it is basic
- **(F)** Baking soda, because it is a base that neutralises the sting

36. The reciprocal of -7/3 is multiplied by -7/3. What is the result? [Rational Numbers]

- **(A)** -1
- **(B)** 1
- **(C)** 49/9
- **(D)** 0
- **(E)** -49/9
- **(F)** 21/21
- **(G)** -3/7

37. In the same sunlight, dry sand became hotter by 18 deg C while a pond of water became hotter by only 6 deg C. How many times as much did the sand temperature rise compared with the water? [Heat]

- **(A)** 2 times
- **(B)** 108 times
- **(C)** 12 times
- **(D)** 24 times
- **(E)** 3 times
- **(F)** 6 times
- **(G)** 1.5 times

38. In a building, the ground floor is taken as 0. A lift starts there, goes up 9 floors, comes down 4 floors, then comes down another 6 floors. Where does it stop? [Integers]

- **(A)** 7th floor below the ground
- **(B)** Ground floor (0)
- **(C)** 5th floor above the ground
- **(D)** 11th floor below the ground
- **(E)** 19th floor above the ground
- **(F)** 1st floor above the ground (+1)
- **(G)** 1st floor below the ground (-1)

39. Factory wastewater is strongly acidic. Why must it be treated with a base before being released into a river? [Acids, Bases & Salts]

- **(A)** A base adds oxygen to the water
- **(B)** A base turns the water into pure drinking water
- **(C)** A base evaporates the harmful water completely
- **(D)** A base neutralises the acid, making the water safe for fish and plants
- **(E)** A base makes the water more acidic, which kills germs
- **(F)** An acid is added to make it basic before release

40. Find the value of 5/6 - 3/8 (how much greater 5/6 is than 3/8). [Rational Numbers]

- **(A)** 11/14
- **(B)** 11/24
- **(C)** 2/24
- **(D)** 13/24
- **(E)** 2/2
- **(F)** 1/2
- **(G)** 8/14

41. A liquid thermometer column is 6 cm long at 0 deg C and rises 0.5 cm for every 10 deg C increase. How long is the column at 60 deg C? [Heat]

- **(A)** 9 cm
- **(B)** 36 cm
- **(C)** 30 cm
- **(D)** 8.5 cm
- **(E)** 7 cm
- **(F)** 9.5 cm
- **(G)** 6.5 cm

42. Find the value of (-50) divided by (-5), minus the product (3 x (-4)). [Integers]

- **(A)** -58
- **(B)** 2
- **(C)** -2
- **(D)** -12
- **(E)** 120
- **(F)** -22
- **(G)** 22

43. Toothpaste is basic in nature. Why is using toothpaste good for your teeth after eating? [Acids, Bases & Salts]

- **(A)** It adds more acid to dissolve food
- **(B)** It turns the saliva into water
- **(C)** It neutralises the acid made by bacteria in the mouth, protecting the teeth
- **(D)** It makes the mouth more acidic to kill bacteria
- **(E)** It works only because it tastes sweet
- **(F)** It makes the teeth dissolve faster
- **(G)** It coats the teeth with a strong acid

44. Find the value of (7 / -9) divided by (-14/27). [Rational Numbers]

- **(A)** 98/243
- **(B)** -2/3
- **(C)** 6/2
- **(D)** 2/3
- **(E)** -3/2
- **(F)** -6/4
- **(G)** 3/2

45. A cup of tea at 80 deg C cools by 12 deg C in the first 5 minutes and by 8 deg C in the next 5 minutes. What is its temperature after 10 minutes? [Heat]

(A) 20 deg C
(B) 72 deg C
(C) 68 deg C
(D) 64 deg C
(E) 76 deg C
(F) 60 deg C

46. Find the value of the product (-1) x (-1) x (-1) x (-1) x (-1) x (-1) x (-1). [Integers]

(A) -1
(B) 7
(C) -6
(D) 0
(E) 1
(F) -7

47. When lemon juice is added to a small amount of baking soda, there is fizzing and the mixture stops being basic. What best describes what happened? [Acids, Bases & Salts]

(A) Two acids reacted together to form a base
(B) The lemon juice turned into a salt by itself
(C) Only water was formed and no gas
(D) A base neutralised another base
(E) An acid (lemon) neutralised a base (baking soda), releasing a gas
(F) Nothing chemical happened; it only mixed
(G) The baking soda made the lemon more acidic

48. Solve for x: x multiplied by (-5/8) equals 1. [Rational Numbers]

(A) 5/8
(B) -40
(C) 40
(D) 8/5
(E) -1
(F) -8/5
(G) -5/8

49. A block of ice is at -6 deg C. It is heated until it reaches its melting point of 0 deg C. By how many degrees must its temperature rise to start melting? [Heat]

(A) 0 deg C
(B) 60 deg C
(C) -12 deg C
(D) 6 deg C
(E) 3 deg C
(F) -6 deg C
(G) 12 deg C

50. In a test of 25 questions, +4 marks are given for a correct answer, -2 for a wrong answer, and 0 for an unattempted one. A student gets 15 correct, 7 wrong, and leaves 3 blank. What is the score? [Integers]

- **(A)** 53
- **(B)** 60
- **(C)** 74
- **(D)** 34
- **(E)** 46
- **(F)** 40

51. A gardener tests the soil with china rose indicator and it turns green. China rose turns green in a base. What should be added to make the soil good for most crops, which prefer near-neutral soil? [Acids, Bases & Salts]

- **(A)** A strong base, to balance the soil
- **(B)** More of a base, to make it greener
- **(C)** An acidic material such as compost or organic matter, to neutralise the base
- **(D)** Lime, because it is an acid
- **(E)** Common salt, because it is basic
- **(F)** Pure water, because it removes the base

52. Find the value of (-5/12) + (-3/8). [Rational Numbers]

- **(A)** -19/24
- **(B)** -2/24
- **(C)** -8/20
- **(D)** -1/24
- **(E)** 19/24
- **(F)** -15/96

53. A nurse reads a clinical thermometer as 39.4 deg C. Normal body temperature is taken as 37 deg C. By how much is the patient temperature above normal? [Heat]

- **(A)** 76.4 deg C
- **(B)** 2 deg C
- **(C)** 2.4 deg C
- **(D)** 3.4 deg C
- **(E)** 2.6 deg C
- **(F)** 24 deg C
- **(G)** 1.4 deg C

54. The temperature in a room falls steadily by 3 deg C every hour, starting from 5 deg C. After how many hours will it reach -13 deg C? [Integers]

- **(A)** 9 hours
- **(B)** 3 hours
- **(C)** 4 hours
- **(D)** 6 hours
- **(E)** 5 hours
- **(F)** 8 hours

55. Which one of these is a base? [Acids, Bases & Salts]

(A) Citric acid (lemon)
(B) Carbonic acid (soda water)
(C) Ascorbic acid
(D) Acetic acid (vinegar)
(E) Hydrochloric acid
(F) Calcium hydroxide (lime water)
(G) Tartaric acid

56. Find the value of (2/3 - 5/6) multiplied by (-4). [Rational Numbers]

(A) -1/6
(B) -2/3
(C) 8/3
(D) -4/3
(E) 1/6
(F) 2/3

57. An oven heats up at a steady 8 deg C per minute, starting from 30 deg C. How many minutes does it take to reach 190 deg C? [Heat]

(A) 24 minutes
(B) 20 minutes
(C) 22 minutes
(D) 16 minutes
(E) 23.75 minutes
(F) 27 minutes
(G) 160 minutes

58. Find the value of (15 - 28) x (-2) + (-9). [Integers]

(A) 43
(B) 35
(C) 5
(D) 17
(E) -17
(F) -35
(G) -43

59. Soap solution feels slippery, tastes bitter, and turns red litmus paper blue. From these clues, soap solution is: [Acids, Bases & Salts]

(A) Both acid and base
(B) An indicator only
(C) Pure water
(D) A neutral substance
(E) A base
(F) A salt with no nature

60. Two pieces of ribbon measure 7/10 m and 3/4 m. What is their total length? [Rational Numbers]

- **(A)** 1 and 1/20 m
- **(B)** 1 and 9/40 m
- **(C)** 10/20 m
- **(D)** 21/40 m
- **(E)** 29/40 m
- **(F)** 1 and 9/20 m
- **(G)** 10/14 m

61. A cold drink at 5 deg C is left on a table in a room at 23 deg C. It warms up by 3 deg C every 4 minutes. How long does it take to reach room temperature? [Heat]

- **(A)** 72 minutes
- **(B)** 24 minutes
- **(C)** 30 minutes
- **(D)** 6 minutes
- **(E)** 18 minutes
- **(F)** 12 minutes
- **(G)** 21 minutes

62. A submarine is at a depth of 250 m below the surface (taken as -250 m). It rises at 40 m per minute. What is its position after 3 minutes? [Integers]

- **(A)** -250 m
- **(B)** -120 m
- **(C)** -90 m
- **(D)** -130 m
- **(E)** 130 m
- **(F)** 370 m
- **(G)** -370 m

63. Among these four samples - sodium hydroxide solution, lemon juice, soap solution, and vinegar - how many are bases? [Acids, Bases & Salts]

- **(A)** 1
- **(B)** 3
- **(C)** 2
- **(D)** 0
- **(E)** 4
- **(F)** 2.5

64. A number multiplied by -3/5 gives 9/20. What is the number? [Rational Numbers]

- **(A)** -3/4
- **(B)** -27/100
- **(C)** 4/3
- **(D)** 27/100
- **(E)** 3/4
- **(F)** -15/4
- **(G)** -4/3

65. Over four days the midday temperatures were 31 deg C, 28 deg C, 35 deg C, and 30 deg C. What was the average midday temperature? [Heat]

- **(A)** 33 deg C
- **(B)** 31.5 deg C
- **(C)** 124 deg C
- **(D)** 32 deg C
- **(E)** 30 deg C
- **(F)** 31 deg C
- **(G)** 29 deg C

66. Find the value of ((-64) divided by 8) x (-3), then divided by (-2). [Integers]

- **(A)** 24
- **(B)** 12
- **(C)** -6
- **(D)** -12
- **(E)** 48
- **(F)** -24
- **(G)** -48

67. Distilled water is neutral. What happens when you test it with both blue and red litmus papers? [Acids, Bases & Salts]

- **(A)** Blue litmus turns red only
- **(B)** Both papers turn red
- **(C)** The papers dissolve
- **(D)** Both papers turn green
- **(E)** Both papers turn blue
- **(F)** Neither paper changes colour
- **(G)** Red litmus turns blue only

68. Find the value of (-7/8) - (5/6) + (1/4). [Rational Numbers]

- **(A)** -7/48
- **(B)** 35/24
- **(C)** -35/24
- **(D)** -19/24
- **(E)** -1/24
- **(F)** -3/8

69. A lab thermometer can read from -10 deg C to 110 deg C. What is the full range (the number of degrees between the lowest and highest readings)? [Heat]

- **(A)** -120 deg C
- **(B)** 10 deg C
- **(C)** 110 deg C
- **(D)** 90 deg C
- **(E)** 100 deg C
- **(F)** 120 deg C

70. Find the value of the sum of all integers from -3 to 6 (including both -3 and 6). [Integers]

- **(A)** 21
- **(B)** -15
- **(C)** 15
- **(D)** 0
- **(E)** 18
- **(F)** 9

71. A solution turns china rose indicator dark pink (the colour it gives in acids). If you test the SAME solution with litmus papers, what should you expect? [Acids, Bases & Salts]

- **(A)** Red litmus turns blue; blue litmus stays blue
- **(B)** Both litmus papers turn green
- **(C)** Blue litmus turns red; red litmus stays red
- **(D)** The solution is a base
- **(E)** Blue litmus stays blue; red turns red
- **(F)** Neither litmus paper changes
- **(G)** Both litmus papers turn blue

72. Find the value of (-3/4) of 48 (that is, (-3/4) multiplied by 48). [Rational Numbers]

- **(A)** 12
- **(B)** -64
- **(C)** 64
- **(D)** 36
- **(E)** -9
- **(F)** -12
- **(G)** -36

73. Why does a metal door handle feel colder to the touch than a wooden one in the same cool room, even though both are at the same temperature? [Heat]

- **(A)** The hand makes metal cold by touching it
- **(B)** Wood gives heat to the hand from inside
- **(C)** Metal absorbs cold from the room and stores it
- **(D)** Metal conducts heat away from the hand faster than wood does
- **(E)** Metal is actually at a lower temperature than wood
- **(F)** Wood is hotter because it makes its own heat
- **(G)** Metal contains cold particles

74. A shopkeeper records his daily result over four days as -120, +200, -150, and +30 (in rupees, where negative means a loss). What is his net result? [Integers]

- **(A)** -90
- **(B)** 500
- **(C)** -40
- **(D)** -500
- **(E)** -260
- **(F)** 40

75. Acid is accidentally spilled on a lab bench. Why is a MILD base like baking soda used to clean it, rather than a strong base? [Acids, Bases & Salts]

- **(A)** A strong base would turn the acid into more acid
- **(B)** A mild base makes the acid stronger
- **(C)** A mild base adds water to dilute everything and nothing else
- **(D)** Only acids can clean up acids
- **(E)** A mild base evaporates the acid instantly
- **(F)** A mild base safely neutralises the acid without itself being dangerous
- **(G)** A strong base is too weak to neutralise acid

76. Find the value of (5/9) divided by (-10/27). [Rational Numbers]

- **(A)** 3/2
- **(B)** 2/3
- **(C)** -2/3
- **(D)** 50/243
- **(E)** -15/2
- **(F)** -50/243
- **(G)** -3/2

77. In a room heater, the air near the heater warms up, becomes lighter, and rises, while cooler air sinks to take its place. What is this circulating movement of heat called? [Heat]

- **(A)** Reflection
- **(B)** Insulation
- **(C)** Convection
- **(D)** Radiation
- **(E)** Evaporation
- **(F)** Conduction
- **(G)** Expansion

78. Find the value of (-7) x 6 + (-7) x 4, and identify the property that makes it equal to (-7) x (6 + 4). [Integers]

- **(A)** -70 (distributive property)
- **(B)** -70 (commutative property)
- **(C)** -2 (distributive property)
- **(D)** -70 (associative property)
- **(E)** 70 (distributive property)
- **(F)** -42 (distributive property)

79. A few drops of an unknown solution turn moist red litmus paper blue. Which substance could it be? [Acids, Bases & Salts]

- **(A)** Distilled water
- **(B)** Dilute hydrochloric acid
- **(C)** Tamarind water
- **(D)** Sodium hydroxide solution (a base)
- **(E)** Soda water (carbonic acid)
- **(F)** Lemon juice
- **(G)** Vinegar

80. Find the value of (-11/6) + (7/4). [Rational Numbers]

- **(A)** -4/2
- **(B)** -3/12
- **(C)** -1/12
- **(D)** 4/2
- **(E)** -18/10
- **(F)** 18/10
- **(G)** 1/12

81. We feel the warmth of the Sun even though space between the Sun and the Earth is almost empty. How does this heat reach us? [Heat]

- **(A)** By sound waves carrying heat
- **(B)** By convection currents in space
- **(C)** By the Sun touching the Earth
- **(D)** Heat cannot travel through empty space
- **(E)** By radiation, which needs no material medium
- **(F)** By conduction through the air in space

82. Find the missing number: 36 divided by (the missing number) equals -4. [Integers]

- **(A)** -144
- **(B)** 144
- **(C)** -9
- **(D)** 4
- **(E)** 9
- **(F)** -4
- **(G)** -32

83. When an acid reacts with a base, heat is given out and a salt and water form. Using this, what would you observe in the test tube during a neutralisation? [Acids, Bases & Salts]

- **(A)** Only a gas is produced and nothing else
- **(B)** The mixture becomes slightly warm as the reaction releases heat
- **(C)** The mixture freezes solid
- **(D)** No change happens at all
- **(E)** The mixture turns into a strong base
- **(F)** The mixture turns into pure acid
- **(G)** The mixture becomes ice-cold

84. Find a rational number that lies exactly halfway between -2/5 and -1/5. [Rational Numbers]

- **(A)** -1/10
- **(B)** -2/10
- **(C)** -1/5
- **(D)** -1/2
- **(E)** -3/10
- **(F)** 3/10

85. A railway track is built from steel rails, and small gaps are left between the rails. Why are these gaps necessary? [Heat]

- **(A)** Because the gaps make the train faster
- **(B)** To save steel and reduce the cost of the track
- **(C)** So the rails have room to expand when they get hot, without buckling
- **(D)** Because steel shrinks when heated and needs space
- **(E)** So that the rails can slide off in winter
- **(F)** So that water can drain through the gaps
- **(G)** So the train wheels can grip the track

86. Find the value of (-2) x (-3) x (-4) x (-5). [Integers]

- **(A)** 60
- **(B)** 14
- **(C)** 120
- **(D)** -60
- **(E)** -14
- **(F)** -120
- **(G)** 24

87. A piece of natural cloth is dyed yellow with turmeric. A drop of soap solution is placed on it, leaving a red spot. What does the red spot tell you about the soap solution? [Acids, Bases & Salts]

- **(A)** It is acidic, because turmeric turns red in an acid
- **(B)** It is pure water
- **(C)** It is an acid that bleaches turmeric
- **(D)** It is neutral, because turmeric never changes colour
- **(E)** It is a salt with no effect
- **(F)** It is basic, because turmeric turns red in a base

88. Write the rational number -18/24 in its standard (lowest) form. [Rational Numbers]

- **(A)** -4/3
- **(B)** -9/12
- **(C)** -3/4
- **(D)** -6/8
- **(E)** 3/4
- **(F)** -2/3

89. A thermos flask keeps drinks hot using a vacuum between two glass walls and shiny silvered surfaces. The vacuum gap mainly prevents heat loss by which methods? [Heat]

- **(A)** Convection only, since heat cannot conduct through glass
- **(B)** Conduction only, since glass cannot convect
- **(C)** Conduction and convection, because there is no air in the vacuum to carry heat
- **(D)** It prevents radiation but freely allows conduction
- **(E)** Evaporation and condensation
- **(F)** It does not prevent any heat loss at all
- **(G)** Radiation only, which the vacuum blocks completely

90. Find the value of (-7) x (the missing number) + 12 = -30. What is the missing number? [Integers]

- **(A)** 42
- **(B)** -3
- **(C)** 3
- **(D)** -42
- **(E)** 6
- **(F)** -6
- **(G)** 18

91. Lemon juice and orange juice taste sour. Curd also tastes sour. What does this sour taste usually indicate about these foods? [Acids, Bases & Salts]

- **(A)** They contain indicators
- **(B)** They contain acids (such as citric or lactic acid)
- **(C)** They are basic because they are foods
- **(D)** They are perfectly neutral
- **(E)** They contain only salts
- **(F)** They contain bases

92. Order these rational numbers from smallest to largest: -1/2, 2/3, -3/4, 1/4. [Rational Numbers]

- **(A)** -3/4, -1/2, 2/3, 1/4
- **(B)** 1/4, -1/2, -3/4, 2/3
- **(C)** -1/2, -3/4, 2/3, 1/4
- **(D)** 2/3, 1/4, -1/2, -3/4
- **(E)** -1/2, -3/4, 1/4, 2/3
- **(F)** 2/3, -3/4, -1/2, 1/4
- **(G)** -3/4, -1/2, 1/4, 2/3

93. On a cold night, the land cools to 18 deg C while the sea stays at 24 deg C. Which is warmer, by how much, and which way does the breeze blow? [Heat]

- **(A)** The sea is warmer by 6 deg C; the breeze blows from sea to land
- **(B)** The sea is warmer by 6 deg C; the breeze blows from land to sea
- **(C)** The sea is warmer by 42 deg C; the breeze blows from land to sea
- **(D)** They are equal; no breeze blows
- **(E)** The land is warmer by 42 deg C; the breeze blows from land to sea
- **(F)** The land is warmer by 6 deg C; the breeze blows from land to sea
- **(G)** The land is warmer by 6 deg C; the breeze blows from sea to land

94. A frozen-food store is at -15 deg C. To defrost it, the temperature is raised by 4 deg C each hour. What is the temperature after 6 hours? [Integers]

- **(A)** -39 deg C
- **(B)** 24 deg C
- **(C)** 39 deg C
- **(D)** 11 deg C
- **(E)** -9 deg C
- **(F)** 9 deg C

95. You are given three solutions and only RED litmus paper to test them. Solution P turns it blue, solution Q does not change it, and solution R does not change it. Which conclusion is definitely correct? [Acids, Bases & Salts]

- **(A)** All three are bases
- **(B)** P is an acid; Q and R are bases
- **(C)** P is a base; Q and R are not bases (they are acidic or neutral)
- **(D)** P, Q and R are all neutral
- **(E)** P is neutral; Q and R are bases
- **(F)** Q and R are definitely acids

96. A water tank is 3/4 full. Then 1/3 of the water in it is used. What fraction of the FULL tank is now left? [Rational Numbers]

- **(A)** 3/12
- **(B)** 1/3
- **(C)** 1/2
- **(D)** 7/12
- **(E)** 5/12
- **(F)** 1/4
- **(G)** 2/3

97. A child wraps a hot tiffin box in a thick woollen cloth to keep the food warm for longer. Which explanation is correct? [Heat]

- **(A)** Wool traps air, and air is a poor conductor, so it slows heat loss from the food
- **(B)** Wool turns the food warm by a chemical reaction
- **(C)** Wool reflects sound to keep the box warm
- **(D)** Wool cools the food slowly by adding cold
- **(E)** Wool is a good conductor that pulls heat into the food
- **(F)** Wool produces heat that keeps the food warm
- **(G)** Wool makes the food heavier so it stays warm

98. Find the value of (-5) x (8 + (-3)) - (-20) divided by 4. [Integers]

- **(A)** -15
- **(B)** -30
- **(C)** -20
- **(D)** -45
- **(E)** -25
- **(F)** 30

99. Which statement correctly matches an indicator with its colour change? [Acids, Bases & Salts]

- **(A)** Red litmus turns green in a base
- **(B)** Blue litmus turns blue in an acid
- **(C)** Red litmus turns red in a base
- **(D)** Litmus has no colour change with acids
- **(E)** Blue litmus turns RED in an acid
- **(F)** China rose turns green in an acid
- **(G)** Turmeric turns red in an acid

100. Find the value of (3/4) divided by (-9/16), and then simplify. [Rational Numbers]

(A) -12/9

(B) -3/4

(C) 27/64

(D) 4/3

(E) 3/4

(F) -27/64

(G) -4/3